	AzurESky	
	AS015IW	PROCEDURE ON THE DETERMINATION OF TOTAL MOISTURE
	Compiled by:	Tshepo Maponya- Laboratory Supervisor
	Approved by:	Nonhlanhla Msibi- Operations Manager

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Date implemented	Revision no	Next rev date
08/2025	00	08/2028

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1. Scope and Principle

This method describes the determination of total moisture in coal using a two-stage or single-stage thermal drying process. Moisture is determined by the loss in mass of the sample when heated in air at 105 °C to 110 °C.

Reference Standards: ISO 589, ISO 13909-4, ISO 18283

2. Sample Preparation and Division

2.1. Primary Split (Bulk Sample)

1. Obtain the primary bulk sample.
2. **Cone and Quartering:** Form the coal into a cone, flatten the top, and divide into four quarters.
    - Select one quarter. If the mass is less than the required **35 kg** for a 50 mm top-size, take two opposite quarters to ensure representativeness.
3. **Initial Mass (m_1):** Immediately weigh the split sample and record the mass on LIMS . This is the "As Received" weight.

2.2. Stage 1: Air Drying (Free Moisture)

1. Spread the 35 kg sample thinly on clean drying pans.
2. Allow the sample to air dry at ambient temperature (or in a fan-forced oven not exceeding 40 °C) until "visibly dry."
3. **Stability Check:** Weigh the pans back on LIMS. Air drying is complete when the mass loss is less than 0.1g.
4. Record the final air-dried mass (m_2).

3. Sub-Sampling and Crushing (ISO 13909-4)

3.1. Crushing Sequence

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- **Primary Crush:** Pass the air-dried sample through the 25 mm crusher.
- **Division:** Use a Riffle Splitter to reduce the mass in between the stage crushing.
- **Secondary Crush:**
- **For Single-Stage:** Crush to **-10 mm**.
    - **For Two-Stage:** Crush to **-3 mm**.

3.2. Final Preparation

1. Riffle the crushed material to achieve a sample of 1kg per riffled side.
2. **Duplicate Pans:** Prepare two clean, dry pans. Tare the pans on the LIMS-connected balance.
3. Transfer the 1 kg sub-sample into the pans and capture the mass on LIMS for both **Pan A** and **Pan B**.

4. Oven Drying (Residual Moisture)

4.1. Two-Stage Moisture (-3 mm)

1. Place pans in the oven at **105 °C to 110 °C**.
2. **Initial Interval:** Dry for **3 hours**.
3. **First Weighing:** Capture mass on LIMS after 3 hours then return to oven for 30 minutes.
4. **Stability Criteria:** If mass loss between the 30-minute intervals is 0.1g, the sample will read “stable” signalling that sample analysis is complete.
5. **Failure Limit:** If the sample does not stabilize within **5 hours**, the test is marked as "Failed" in LIMS and must be repeated.

4.2. Single-Stage Moisture (-10 mm)

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1. Follow the same procedure as above, but the sample remains in the oven for the full **5-hour duration** without the 3-hour initial lapse check.
2. Capture final stabilized mass on LIMS.

5. Calculations

5.1. Surface (Free) Moisture (M_ad)

$m_{ar} = \frac{m_2 - m_1}{m_1}$ x 100

Where:

- m_1 = Initial mass of sample as received.
- m_2 = Mass of sample after air drying.

5.2. Residual Moisture (M_res)

$$M_{res} = \frac{m_4 - m_3}{m_3} X\ 100$$

Where:

- m_3 = Mass of sample before oven drying (1 kg).
- m_4 = Mass of sample after oven drying (stabilized).

5.3. Total Moisture (TM)

For a two-stage determination, the moisture is calculated as follows:

$$TM = m_{ar} + \frac{(100 - m_{ar})\ X\ m_{res}}{100}$$

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6. Precision Limits (ISO 589)

Note: If the difference between Pan A and Pan B exceeds 0.30, the LIMS should trigger a "repeat" alert, requiring a repeat of the sample.

Property	Repeatability	Reproducibility
Second Stage moisture	0.3	Not Applicable
Total moisture	0.5	1.5

Note: LIMS calculates the total moisture slightly differently, wherein it takes the average of the residual moistures and it inserts it directly into the TM formula, whereas the standard calculates the total moisture of the pans & averages them at the end of the calculation. However, both methods are accepted.

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